



Sexier HDDs

Western Digital's external HDDs are receiving a makeover on the outside. <http://dgit.in/SexierHDD>



Setting up Phillips Hue

Phillips smart lighting from the Hue series can be controlled right from your iPhone to take its full advantage. <http://dgit.in/PhlpsHue>

From the labs

The DNA containing the gene had to be stitched onto the patient's DNA at the right place. Mistakes happen, and the DNA could get stitched in at the wrong site, because it bore a resemblance to the target site. Plus, when gene integration goes wrong, it could lead to serious complications such as cancer. It's not just about safety though, because even efficiency is a concern. This is because for some diseases, more than one gene needs to be replaced, which poses a problem for virus mediated gene delivery systems. This is why geneticists have been on a constant lookout for better techniques. CRISPR, machinery borrowed from tiny bacteria, offers a new way forward.

CRISPR: Bacteria's self defense mechanism

As is usually the case, solutions pop up when no one is looking for one. Believe it or not, CRISPR was discovered by accident because the yogurt industry wanted their good bacteria to be healthier. They wanted bacteria that would be more resistant to viruses that were infecting the bacteria and spoiling the yogurt. While looking at virus resistant bacteria, scientists found the CRISPR mechanism.

CRISPR is something bacteria have been using for ages to fight against viruses. In a region of the bacteria's DNA, there's an area called CRISPR. This region has a whole host of Short Palindromic Repeats (short DNA sequences that read the same forwards as well as backwards). The sequences are interspaced at regular intervals, and in clusters, which is why we arrive at the name: Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR). In the spaces between the Short Palindromic Repeats, the bacteria store DNA identities of viruses that have attacked it before. This spacer DNA is actually where all the magic happens.

In addition to the CRISPR section, scientists also discovered CRISPR-associated (Cas) genes (Cas9 for this story), which work with CRISPR to defend the bacteria against infection.

These Cas genes have been assigned numbers as a way to identify them. The Cas genes make Cas proteins, which are either nucleases or helicases. Helicases are enzymes that unravel DNA, and nucleases are those that can cut DNA strands.

If the bacteria is infected by a virus DNA that it has already encountered, the Cas genes order up some Cas proteins, the CRISPR chunk identifies the viral DNA and orders the creation of a CRISPR-RNA (which identifies the viral DNA and bonds with it, while the nuclease cuts up the viral DNA). If not for CRISPR, the viral DNA would latch on somewhere, replicate and quickly kill off the cell. Instead, CRISPR neutralises the threat.

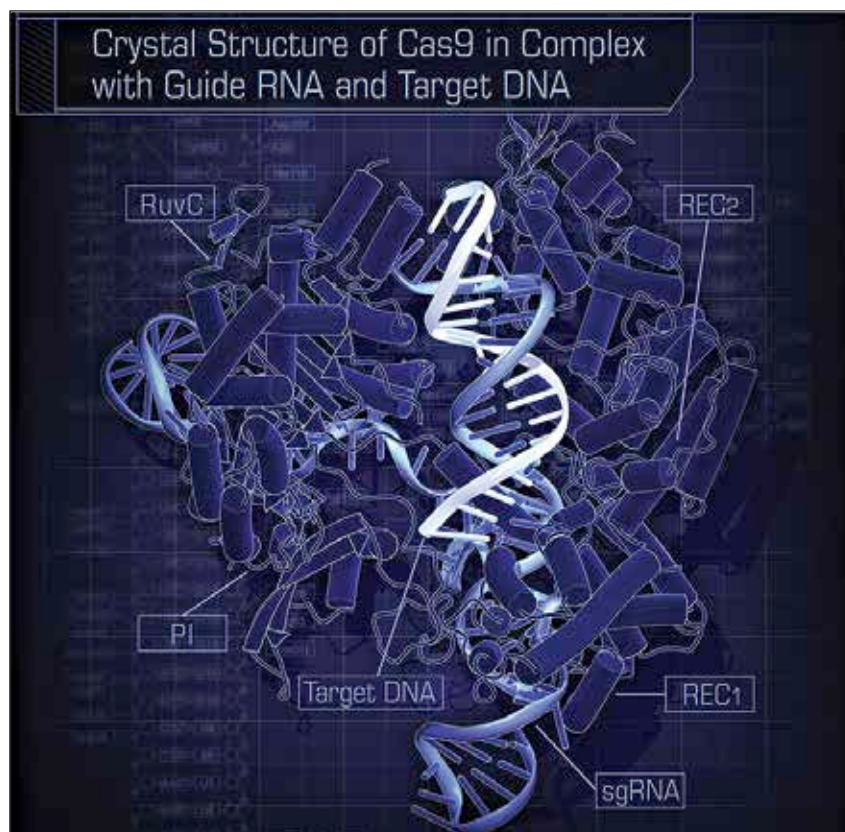
But what if it's a new infection that's not in CRISPR's record? Here too CRISPR works, because it creates a Cas protein, which will then tackle the invader, and then break it's DNA apart.

Geneticists borrowed the nuclease and customised the guide RNA so that it could be used to alter human DNA.

How it works

For certain diseases, we know how and where the DNA code has gone wrong and we can pinpoint the faulty sequence. To rectify the problem, the editing tool needs to first find that sequence in our DNA. This is like looking for a needle in a stack of needles! Here's where the guide RNA helps us. In bacteria, the guide RNA, coming from the archival CRISPR region, finds the sequence with ease, however, in our case, because we know the sequence that needs to be edited, we don't need the archive. Instead, we synthesize the RNA guide ourselves (because we know the sequence) and put it inside human cells along with the Cas9 nuclease. Now instead of recognising foreign DNA (as it did in bacteria), the CRISPR machinery recognises the faulty gene in our cells and cuts it out.

Two things can happen when the guide sequence directs the scissor which then cuts the DNA. Either only a few letters in the DNA sequence can be modified on it's own or alternately,



Cartoon showing the guide RNA – Cas9 latching on the target DNA